HIGH PERFORMANCE CONTROLLER FOR APPLICATION PROCESSING & REAL-TIME CONTROL

RZ/T2N INTRODUCTION

RENESAS ELECTRONICS CORPORATION
EMBEDDED PROCESSING PRODUCT GROUP
EMBEDDED PROCESSING 1ST BUSINESS DIVISION

EP1AP-AA-24-0068-01

Subject to NDA

This product is under development.
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  - RZ Family, Roadmap, Portfolio, Lineup, Target Application, Part Number
- Product overview
  - Functional Block
  - Application Use Case (PLC, Servo drive, Robot, Network Card)
- Features
  - System architecture
  - Multi-axis servo control
  - Industrial network
- Development Environment
  - FSP
  - Linux (Verified Linux Package, Multi-OS package, Industrial network package, etc)
  - Evaluation Board

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RENESAS RZ FAMILY

HIGH-PERFORMANCE APPLICATION FOCUSED AND GENERAL-PURPOSE PROCESSORS

Vision AI

RZ/V Series

AI accelerator + Linux

Industrial Network

RZ/N Series

Multi-protocol Industrial Ethernet with Redundancy + Linux/ RTOS

Industrial real-time control

RZ/T Series

Real-time control + Linux/RTOS

RENESAS RZ

Human Machine Interface & IoT Edge

RZ/G Series

3D Graphics + Linux/Android™

RZ/A Series

2D Graphics + RTOS

Vision AI	Renesas own cutting-edge AI technology "DRP-AI" for embedded AI application
Network	TSN and Multi-protocol Industrial network support
Real-time	High-performance Processing and High Precision Realtime Control
HMI	High-performance CPU operation and Graphics/Video Processing
Edge IoT	High-performance CPU and High-speed interfaces for IoT Edge products

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PRODUCT ROADMAP FOR RZ/T AND RZ/N SERIES

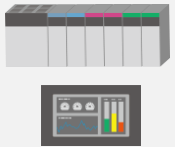
ENHANCING PRODUCT PORTFOLIO TO COVER CUSTOMER'S VARIOUS APPLICATIONS

: MP or Under Development
: Under Planning

489-pin WS: 2025 May
489-pin MP: 2026/1Q

Today's focus

PLC / Controller



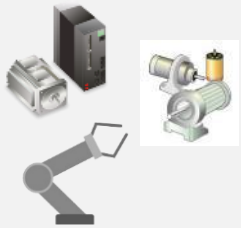
RZ/N1D
CA7x2 (500MHz)
CM3 (125MHz)

RZ/N2H
CA55x4 (1.2GHz)+CR52x2 (1GHz)

Long-Term Linux support, TSN, 5G network, OPC UA & FuSa, Security Enhancement, Compatibility w/ RZ/Tx

RZ/Nxx

Servo / Inverter / Industrial Robot



RZ/T1
CR4 (600MHz)
CM3 (150MHz)

RZ/T2M
CR52x2 (800MHz)

High-performance CPU TSN & Multi protocol support, FuSa support

RZ/T2L
CR52 (800MHz)

Low-end variant for EtherCAT w/ high performance CPU & FuSa

RZ/T2H
CA55x4 (1.2GHz)+CR52x2 (1GHz)

High-performance Linux CPU & real-time control CPU

RZ/T2ME

Enhanced version of RZ/T2M OTFD, UDP/IPv4 E2E TC support

Under Dev.

RZ/T2N
CA55x2 (1GHz)+CR52x2 (800MHz)

Shrink version of RZ/T2H. Dual CA core + Dual CR core

Remote IO / Network module

RZ/N1S
CA7 (500MHz)
CM3 (125MHz)

RZ/N1L
CM3 (125MHz)

RZ/N2L
CR52 (400MHz)

TSN & PROFINET IRT network controller. Motor control for AC drives.

2022 | 2023 | 2024 | 2025 | 2026~

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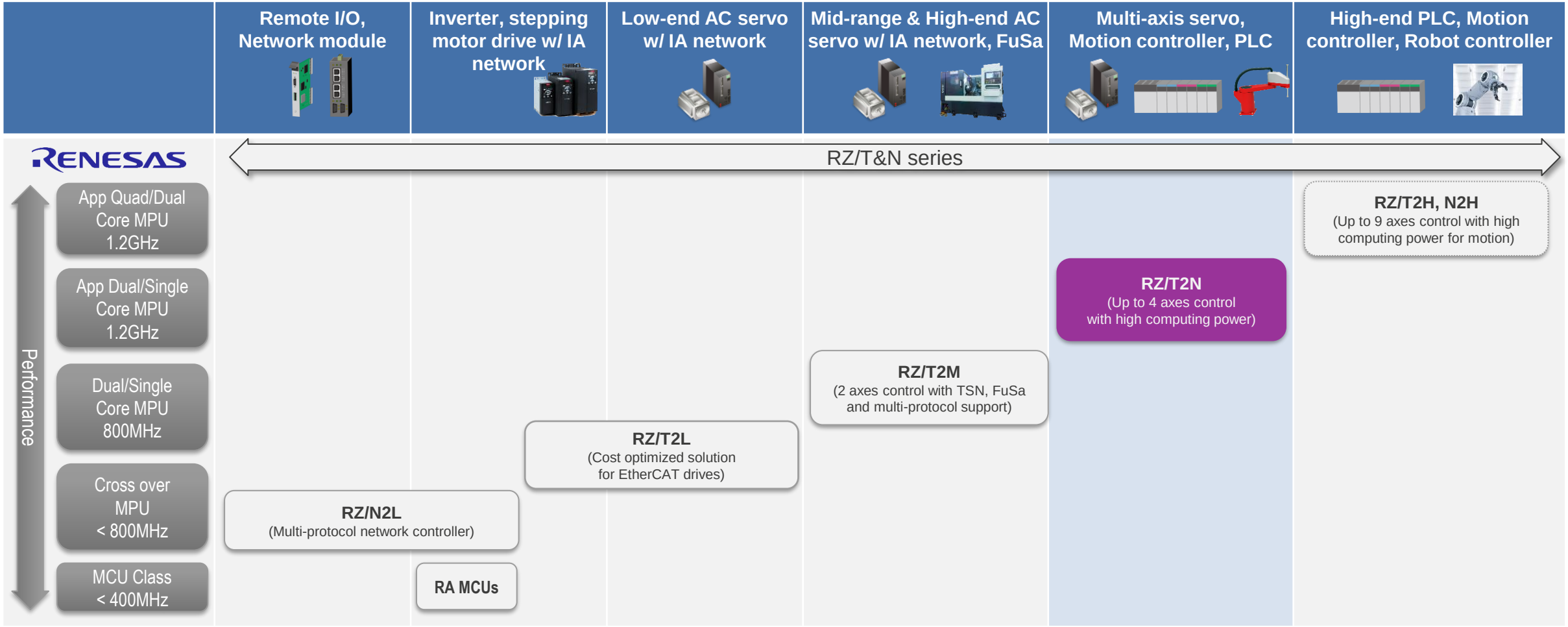
PRODUCT PORTFOLIO FOR RZ/T AND RZ/N SERIES

SCALABLE PRODUCTS TO COVER WHOLE IA SYSTEM



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PRODUCT LINEUP RZ/T2X, N2X

Product concept *IE: Industrial Ethernet	*Linux & Real-time control w/ master/slave IE**		*Real-time control w/ & w/o slave IE*	*Optimized for slave IE*
	RZ/T2H RZ/N2H	RZ/T2N	RZ/T2M RZ/T2L	RZ/N2L
# of Arm® Cortex®-A55	1.2GHz Single/Dual/Quad	1.2GHz Single/Dual	NA	NA
# of Arm® Cortex®-R52	1.0GHz, Dual	800MHz, Dual	800MHz, Single/Dual	400MHz, Single
Industrial Ethernet	Master/Slave Multi-Protocol	Master/Slave Multi-Protocol	Slave Multi-Protocol or EtherCAT or w/o	Slave Multi-Protocol
Max drive motor axis	9-axis for T2H 4-axis for N2H	4-axis	2-axis	2-axis
Absolute Encoder	16ch	8ch	2ch	NA
DDR	LPDDR4-3200 32bit	LPDDR4-1600 16bit	NA	NA
PCIe	2ch Gen3	1ch Gen2	NA	NA
HMI	Yes	NA	NA	NA
PKG	FCBGA729 FCBGA576	FBGA537 FBGA489	FBGA320/225/196 LQFP176/128	FBGA225/121

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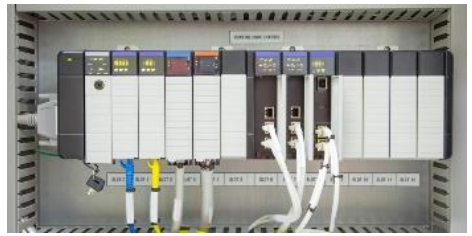
TARGET APPLICATION

RZ/T2N can be adopted to various industrial equipment which Industrial Ethernet capability.



PLC / Motion Controller

AC Servo

SCARA robot

Articulated robot

Collaborative robot

AGV/AMR

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RZ/T2N GROUP PART NUMBER

Series	Group	Security	Part number	Cortex-A55	Cortex-R52	Other option
RZ/T	RZ/T2N (537 FBGA)	Secure	R9A07G076M48GBG	Dual cores	Two CPUs	EtherCAT, CAN-FD, HSR
			R9A07G076M47GBG	Dual cores	Two CPUs	EtherCAT, CAN-FD
			R9A07G076M45GBG	Dual cores	Two CPUs	Classical CAN
			R9A07G076M37GBG	Single core	Two CPUs	EtherCAT, CAN-FD
			R9A07G076M35GBG	Single core	Two CPUs	Classical CAN
		Non-Secure	R9A07G076M43GBG	Dual cores	Two CPUs	EtherCAT, CAN-FD, HSR
			R9A07G076M30GBG	Single core	Two CPUs	Classical CAN
	RZ/T2N (489 FBGA)	Secure	R9A07G076M48GBA	Dual cores	Two CPUs	EtherCAT, CAN-FD, HSR
			R9A07G076M47GBA	Dual cores	Two CPUs	EtherCAT, CAN-FD
			R9A07G076M45GBA	Dual cores	Two CPUs	Classical CAN
			R9A07G076M37GBA	Single core	Two CPUs	EtherCAT, CAN-FD
			R9A07G076M35GBA	Single core	Two CPUs	Classical CAN
		Non-Secure	R9A07G076M43GBG	Dual cores	Two CPUs	EtherCAT, CAN-FD, HSR
			R9A07G076M30GBG	Single core	Two CPUs	Classical CAN

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RZ/T2N GROUP PART NUMBER

R9A 07 G 07 6 M xx G Bx #xC0

- R9A 07 G 07 — Fixed
- 6 — Product group
- M — Sub group
- xx — Option
- G — Quality grade
- Bx — Package
- #xC0 — Qualification level

Product group

7	High-end
6	Secondary ID for High-end
5	Mid-range
4	Low-end

Sub group

M	2nd Generation

Option

Cortex-A55 dual, R52 dual cores		Cortex-A55 single, R52 dual cores	
49	(RFU)	39	(RFU)
48	CANFD, EtherCAT, HSR, Secure	38	(RFU)
47	CANFD, EtherCAT, Secure	37	CANFD, EtherCAT, Secure
46	(RFU)	36	(RFU)
45	Classical CAN, Secure	35	Classical CAN, Secure
44	(RFU)	34	(RFU)
43	CANFD, EtherCAT, HSR, non-Secure	33	(RFU)
42	(RFU)	32	(RFU)
41	(RFU)	31	(RFU)
40	(RFU)	30	Classical CAN, non-Secure

Quality

G	Standard Q2A

Package

BG	537 BGA
BA	489 BGA

Qualification level

#AC0	Tray, bulk
#BC0	Tray, full carton

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PRODUCT OVERVIEW

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RZ/T2N

MID-RANGE MPU FOR HIGH PROCESSING POWER & REAL-TIME CONTROL

Samples	2Q25
Production	3Q26

[Package]
- 21x21mm, 0.8mm pitch, FBGA 537pin
- 16x16mm, 0.65mm pitch, FBGA 489pin

Features

- 2x Cortex-A55 (1.2GHz), 2x Cortex-R52 (800MHz)
- Motor Control Peripherals for up to 4 axes
- Encoder I/F(A-format, BiSS-C, EnDat2.2, HIPERFACE DSL, etc.)
- Industrial Ethernet(TSN, EtherCAT, PROFINET RT/IRT, EtherNet/IP, etc.)
- 16-bit LPDDR4-1600 (in line ECC)
- 1x PCIe (Gen2)
- Functional Safety support

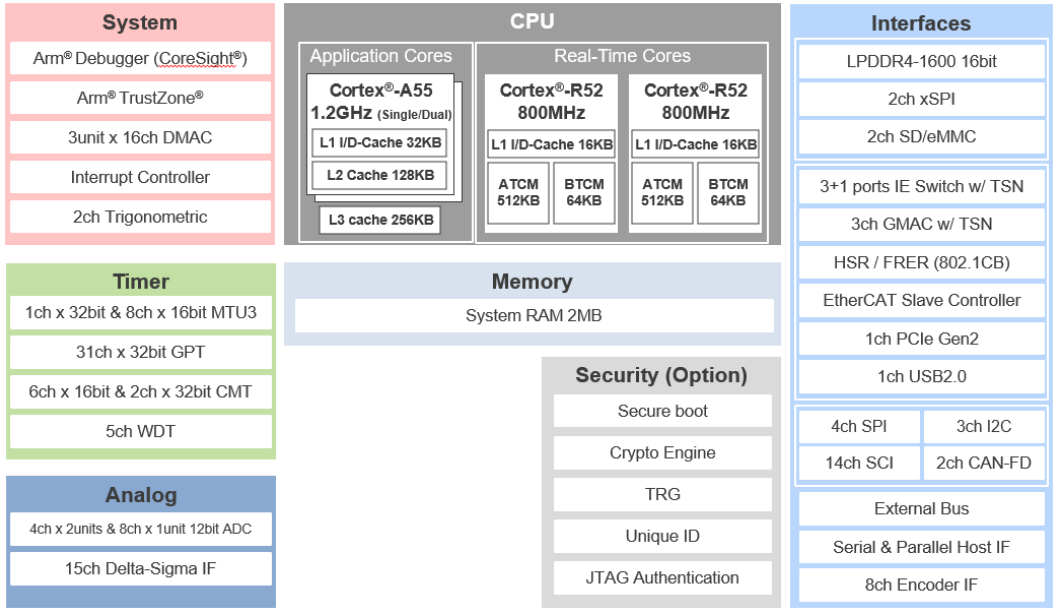
Benefits

- Mid-end one chip solution integrated with processing power for Linux, real-time control, multi-protocol industrial Ethernet and functional safety to dramatically reduces BOM count.
- High real-time performance and motor control peripherals enable up to 4 axes motor control.

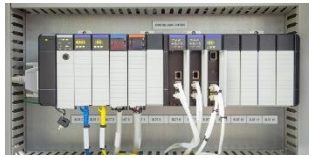
Typical Applications

- PLC, CNC
- Motion controller
- Up to 4 axes motor control such as SCARA robot and ordinary robot
- High-end servo

Block Diagram



Example Application



PLC

SCARA robot

High-end servo

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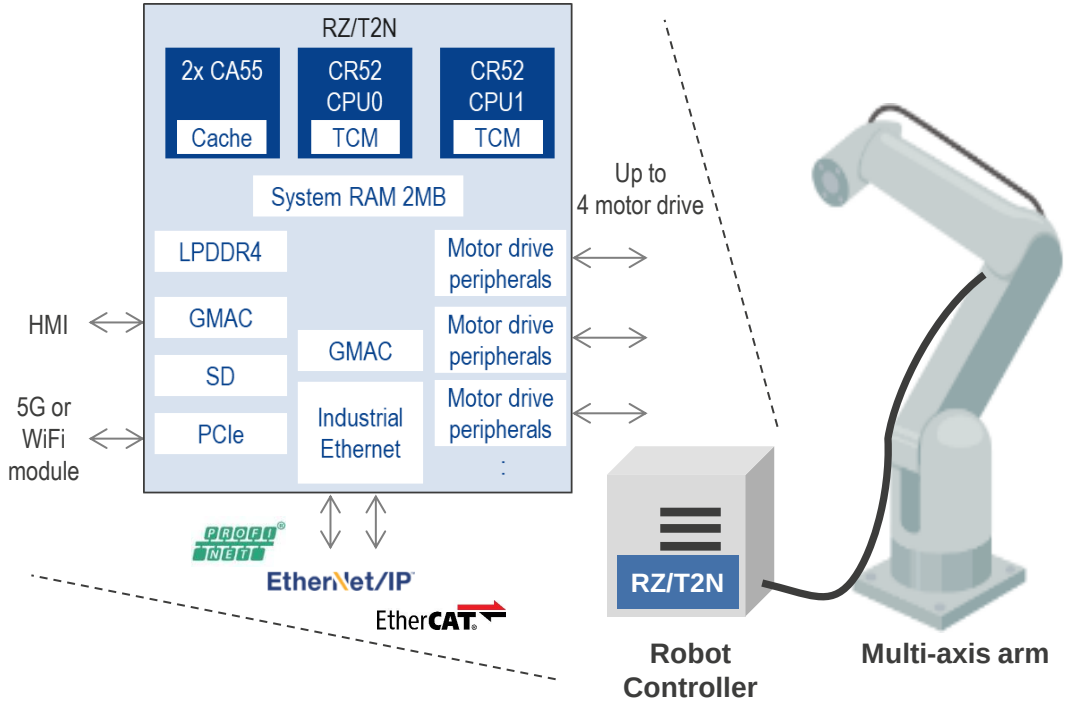
APPLICATION USE CASE
INDUSTRIAL ROBOT

Selectable solution from various products to fit to customer's requirements and system configurations.

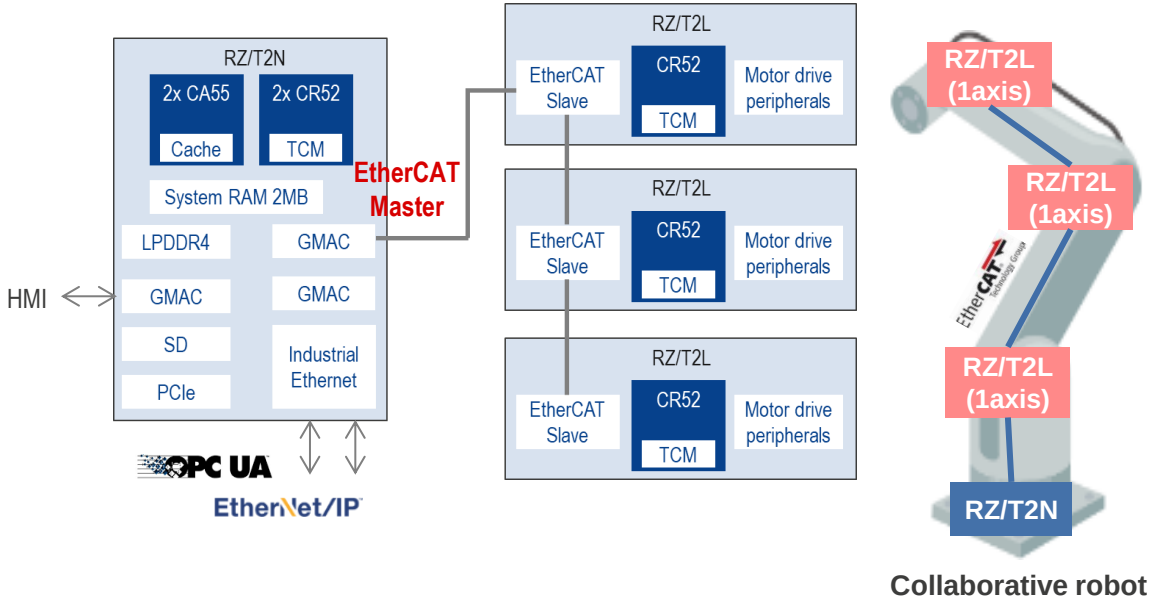
One chip implementation
(ex. Industrial Robot)

RZ/T2N realize motion control and motor control in one chip



Two chip implementation
(ex. Collaborative robot)

RZ/T2N realize motion control and Network master
RZ/T2L realize each axis control



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RZ/T2N FEATURES

SYSTEM ARCHITECTURE

MULTI-AXIS MOTOR CONTROL

INDUSTRIAL NETWORK

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SYSTEM ARCHITECTURE
INTEGRATES KEY FUNCTIONS

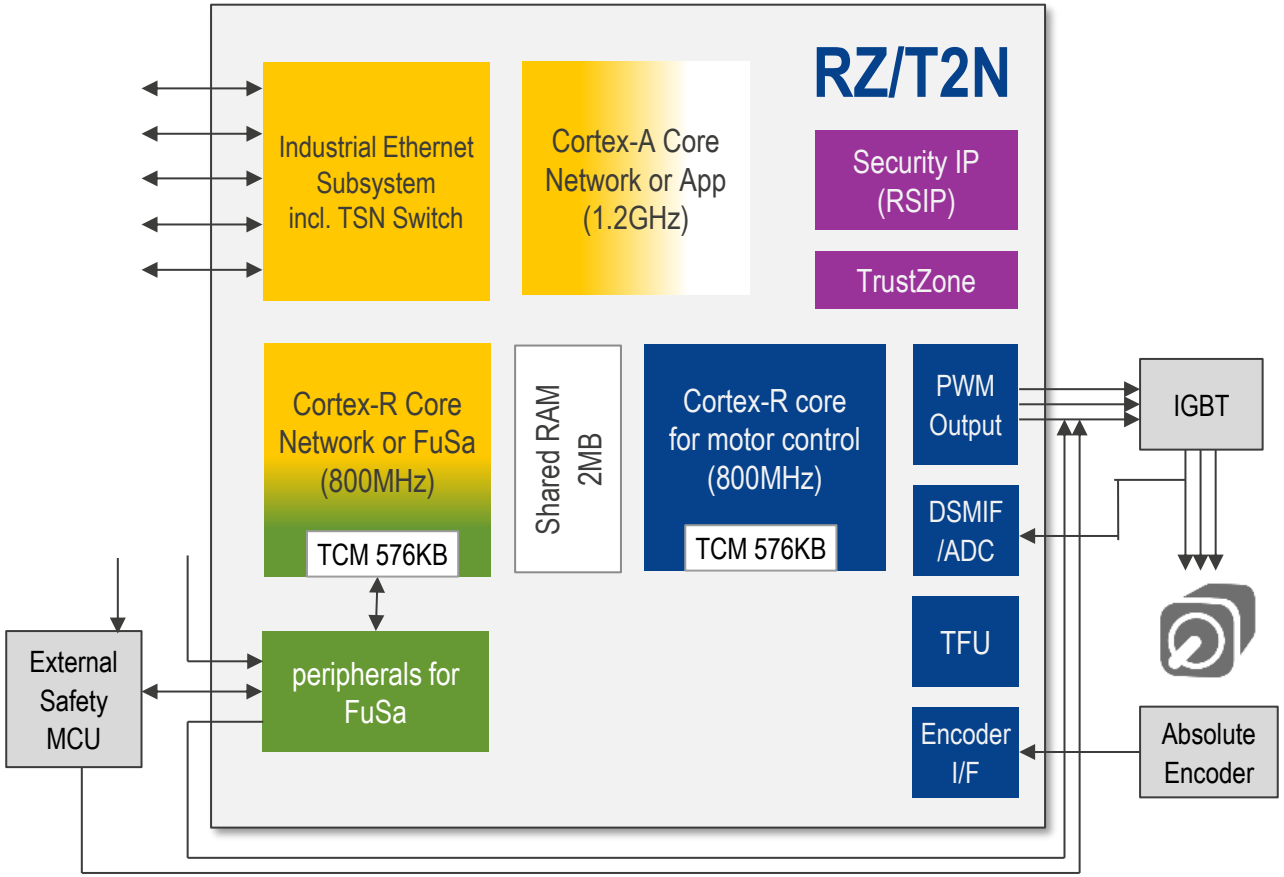
Motor control, Industrial Ethernet, functional safety, and security are all integrated

Industrial Ethernet

- ✓ Gigabit Ethernet up to 5-ports
- ✓ Integrated TSN Switch
- ✓ Multi-protocol support

Functional Safety

- ✓ Internal memory with ECC
- ✓ 1 of 2 external safety MCU can be eliminated
- ✓ Optimized HW configuration for functional safety



Security

- ✓ Secure IP provides high secure key management
- ✓ Secure Boot, Secure Storage, Secure Firmware Update
- ✓ TrustZone

Motor Control

- ✓ 800MHz High performance real-time CPU w/ TCM (576KB)
- ✓ System configuration realizes high precision motor control

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SYSTEM ARCHITECTURE

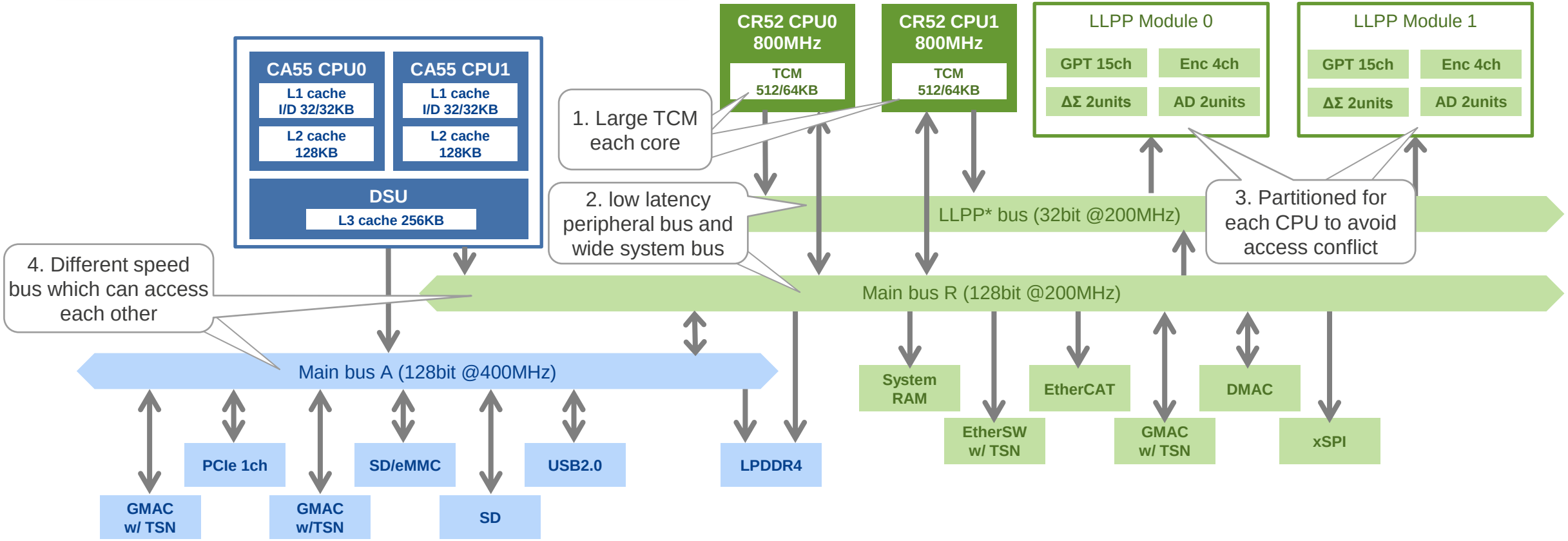
EFFICIENT DOMAIN / BUS SEPARATION

High-performance computing domain

Offers high-performance computing power
by **Quad CA55 CPU@1.2GHz** and **high-speed I/Fs**
w/ **Linux support** for various application

Realtime control domain

Realize high-performance real-time control by **Dual CR52 CPU@800MHz**
w/ **large size TCM** and **motor control peripherals on LLPP*1**
and **Industrial Ethernet slave controllers**



* LLPP : Low Latency Peripheral port

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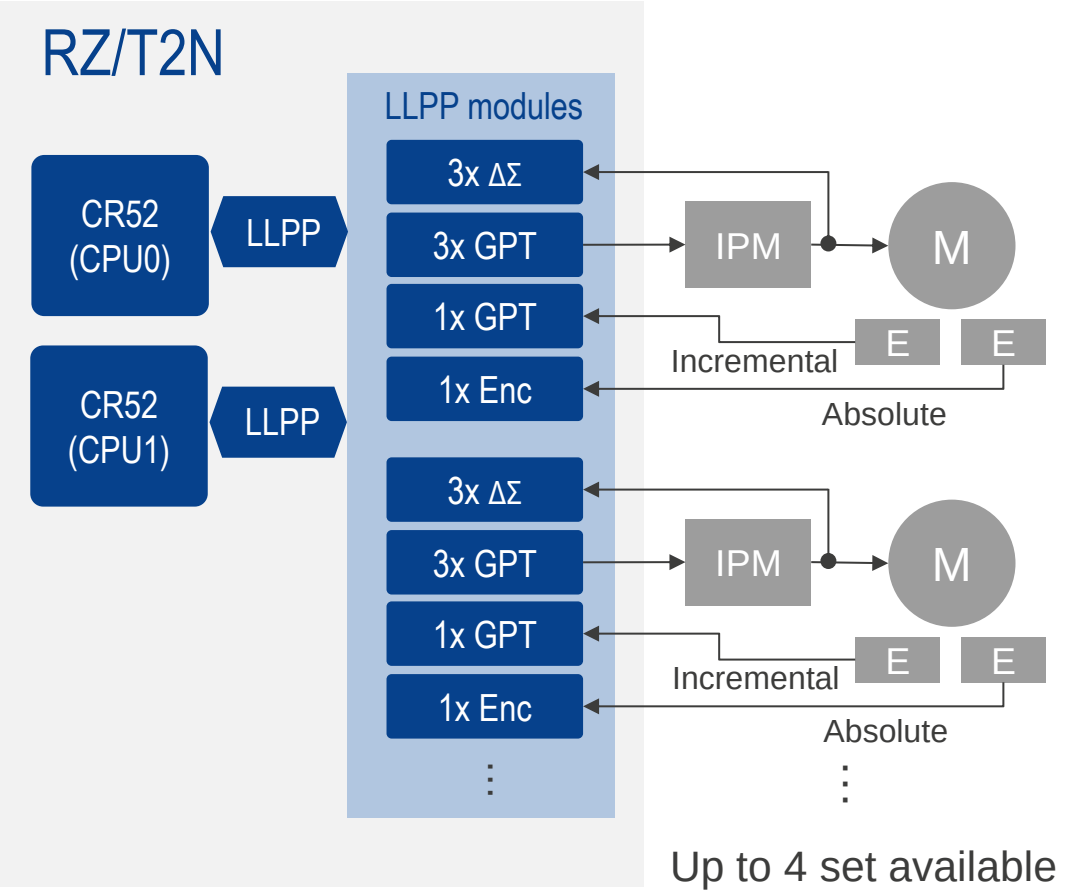
MULTI-AXIS MOTOR CONTROL

4-AXIS MOTOR CONTROL

Up to 4-axis motor drive by peripherals in LLPP modules

- **31ch GPT**
  - ✓ 12ch: PWM output 3ch x 4-axis
  - ✓ 8ch: Incremental encoder: 2ch (speed & position) x 4-axis
  - ✓ 11ch: Other usage (e.g., generic timer, for functional safety)
- **15ch ΔΣ interface**
  - ✓ 12ch: current measurement 3ch x 4-axis
  - ✓ 3ch: Other usage (e.g., voltage monitor)
- **8ch Absolute encoder interface**
  - ✓ Support protocols: **A-format V3**, Tamagawa, Hiperface DSL, BiSS-C, **BiSS-Line**, Endat2.2, **Endat3**
  - ✓ Different protocols are available in parallel each channel

Two encoder use case example
Incremental encoder (speed) & Absolute encoder (position)



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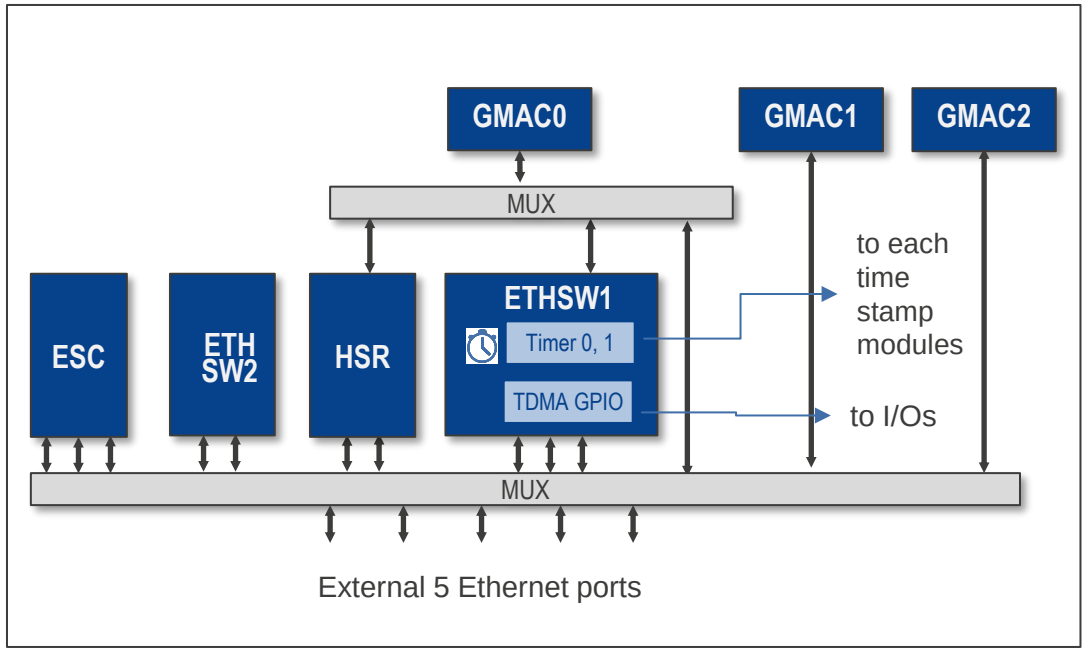
INDUSTRIAL NETWORK

INTEGRATED SWITCH SUPPORTING INDUSTRIAL PROTOCOL

1 Various Configuration Ethernet Subsystem

Integrated multiple switches to cover wide use cases

Realize various protocol configuration with multiplexer



2 Two Gbit TSN switch and Three Gbit TSN Ethernet MAC

Hardware supports TSN standards: IEEE 802.1AS, IEEE 802.1Qbv, IEEE 802.1Qbu, IEEE 802.1Qci, IEEE 802.1Qch, IEEE 802.1CB

Ethernet MAC support TSN end-station, switch supports TSN bridge

3 Support Major Industrial Protocols

Profinet RT/IRT, EtherNet/IP w/ DLR, EtherCAT, OPC UA, TSN

4 Mechanism for Distributed Synchronous Control

Various peripheral hardware can be synchronized based on TSN

Facilitate the realization of Distributed Synchronous Control

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INDUSTRIAL NETWORK
MULTIPLEXING ETHERNET SWITCH & MAC

Multiplexed Ethernet subsystem for main use and extended use

#	ETH0	ETH1	ETH2	ETH3	ETH4
0	ETHSW1			GMAC1	GMAC2
1	ESC			GMAC1	GMAC2
2	HSR		-	GMAC1	GMAC2
3	ETHSW2		GMAC0	GMAC1	GMAC2
4	ETHSW1		GMAC2	HSR	
5	ETHSW1		GMAC2	ETHSW2	
6	ESC		GMAC2	ETHSW2	
7	GMAC0	GMAC1	GMAC2	-	-
8	ETHSW1	ESC	ETHSW1	GMAC1	GMAC2
9	-	ESC		GMAC1	GMAC2

1. ETH0, ETH1 and ETH2
Main use
Industrial Ethernet slave
Industrial Ethernet master
TSN bridge & end-station

2. ETH3 and ETH4
5-ports extended use
Dual switch
Switch and two or three MACs

ETHSW1: Industrial Ethernet switch
ETHSW2: TSN switch supporting FRER
ESC: EtherCAT SubDevice Controller switch
HSR: High-availability Seamless Redundancy switch
GMAC0/1/2: Gigabit Ethernet MAC

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DEVELOPMENT ENVIRONMENT

RTOS / FSP

LINUX

EVALUATION BOARD

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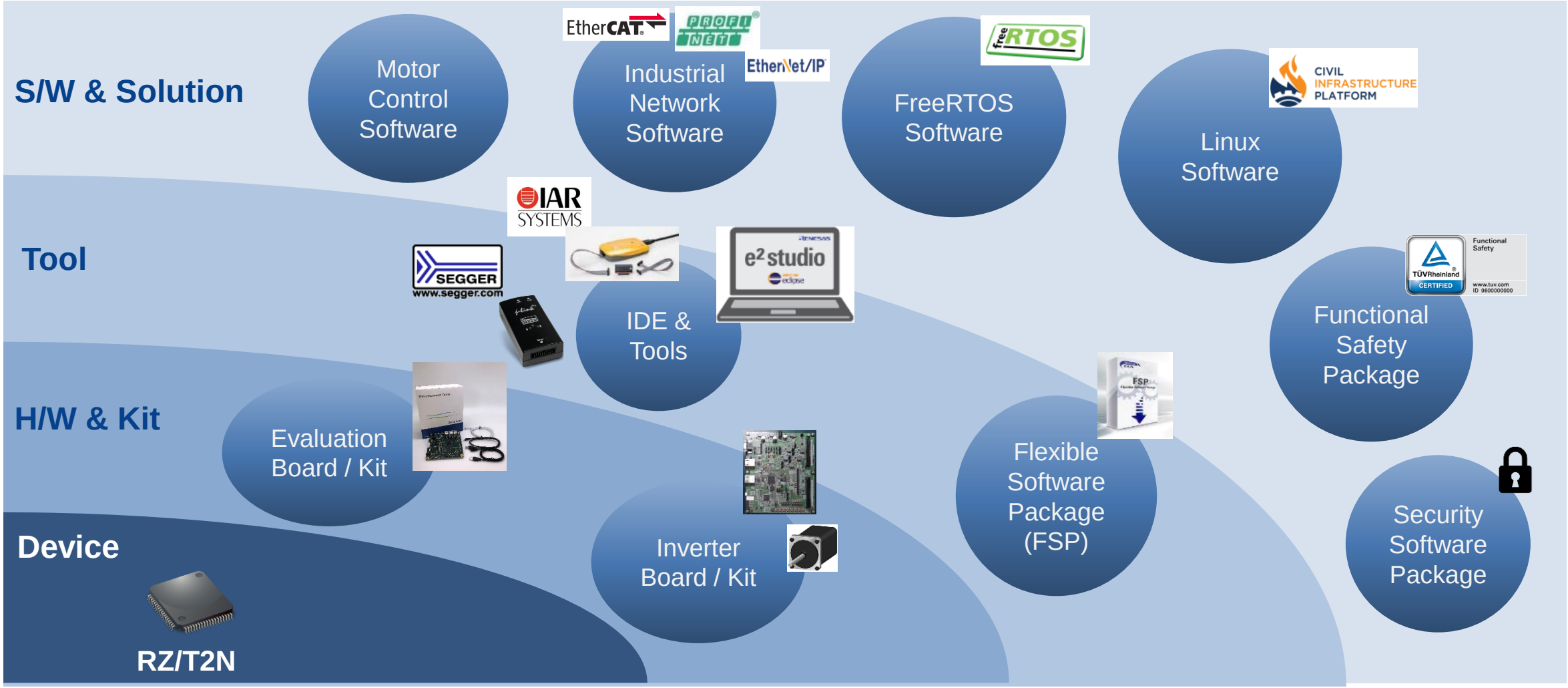
DEVELOPMENT ENVIRONMENT

RENESAS PROVIDES TOTAL SOLUTION



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SCALABLE & COMPATIBLE S/W PLATFORM

ENHANCE REUSABILITY OF CUSTOMER'S S/W ASSETS

- **Flexible Software Platform (FSP)** and **Verified Linux Package (VLP)** bring benefit of compatibility and scalability among Renesas MPU/MCU products to customers

Flexible Software Platform (FSP)

Common S/W platform for Renesas MPU and MCU, which provides configurability by GUI tool.

Verified Linux Package (VLP)

Linux S/W platform based on CIP Linux and common S/W libraries, which are verified by Renesas.

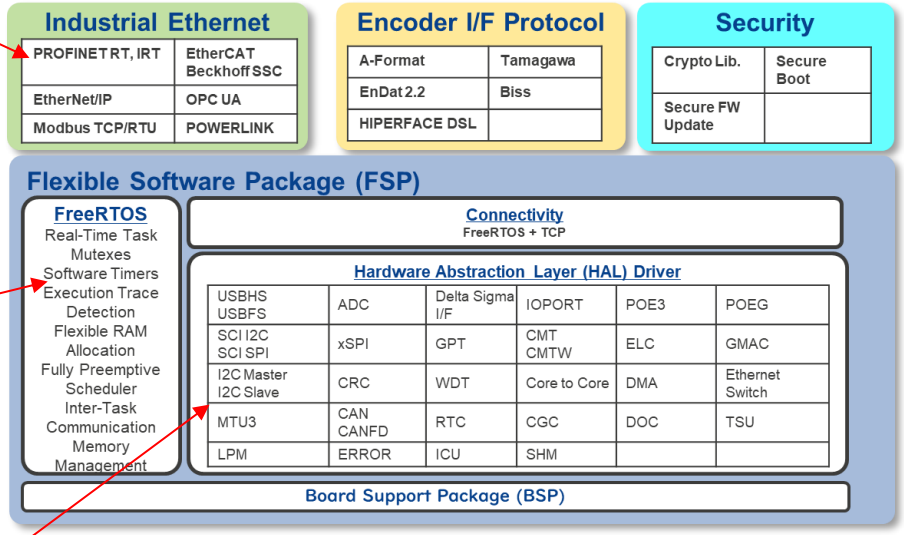
- **Industrial protocol**

EtherNet/IP EtherCAT Modbus DeviceNet PROFINET PROFIBUS CC-Link IE TSN CANopen

- **Realtime OS**

FreeRTOS

- **FSP BSP & HAL drivers**
  - Support GUI configuration by e2Studio



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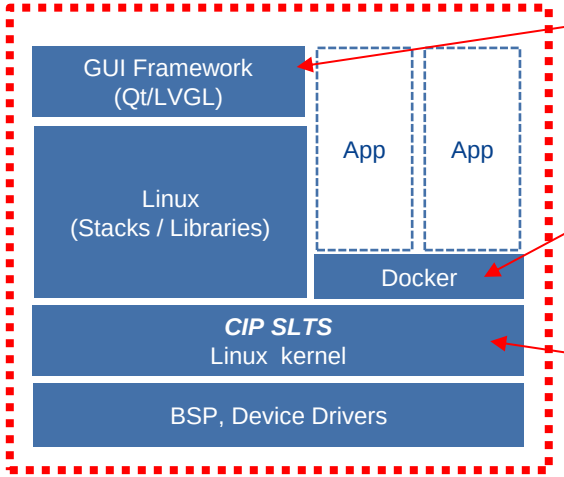


- **GUI framework**

Qt LVGL

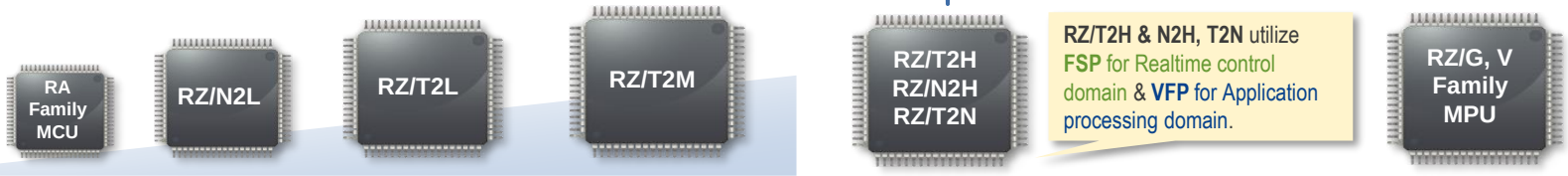
- **Container**

docker

- **Industrial Grade Linux**

CIVIL INFRASTRUCTURE PLATFORM

* CIP SLTS : Industrial grade Linux kernel w/ SLTS (Super Long-Term Support : 10 years)



RZ/T2H & N2H, T2N utilize **FSP** for Realtime control domain & **VFP** for Application processing domain.

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DEVELOPMENT ENVIRONMENT

RTOS / FSP

LINUX

EVALUATION BOARD

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RTOS / FSP (Flexible Software Package)

CONTENTS OVEVIEW

- **RA family like** software structure, **both Cortex-R and Cortex-A** are supported
- **Industrial protocol package** (industrial ethernet and encoder protocol) are also available

Industrial Ethernet

PROFINET RT, IRT	EtherCAT Beckhoff SSC
EtherNet/IP	OPC UA
Modbus TCP/RTU	POWERLINK

Encoder I/F Protocol

A-Format	FA-CODER
EnDat 2.2	Biss
HIPERFACE DSL	

Security

Crypto Lib.	Secure Boot
Secure FW Update	

Flexible Software Package (FSP)

FreeRTOS

Real-Time Task
Mutexes
Software Timers
Execution Trace Detection
Flexible RAM Allocation
Fully Preemptive Scheduler
Inter-Task Communication
Memory Management

Connectivity
FreeRTOS + TCP

Hardware Abstraction Layer (HAL) Driver

USBHS USBFS	ADC	Delta Sigma I/F	IOPORT	POE3	POEG
SCI I2C SCI SPI	xSPI	GPT	CMT CMTW	ELC	GMAC
I2C Master I2C Slave	CRC	WDT	Core to Core	DMA	Ethernet Switch
MTU3	CAN CANFD	RTC	CGC	DOC	TSU
LPM	ERROR	ICU	SHM		

Board Support Package (BSP)

- BSP for evaluation board
- HAL Drivers for peripherals
- FreeRTOS support
- FreeRTOS + TCP
- USB & Ethernet stack
- Software/Hardware security
- Industrial network protocol
- FreeRTOS + FAT file system

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RTOS / FSP (Flexible Software Package)
TOOL CHAIN

Support General MCU tool chain

	IAR EWARM	e² studio
IDE / Debugger		
Compiler	IAR C/C++ Compiler	GCC
Emulator	IAR I-jet IAR I-jet Trace	Segger J-Link
Support tool	Smart Configurator (Stand-alone)	Smart Configurator (e² studio Integration)

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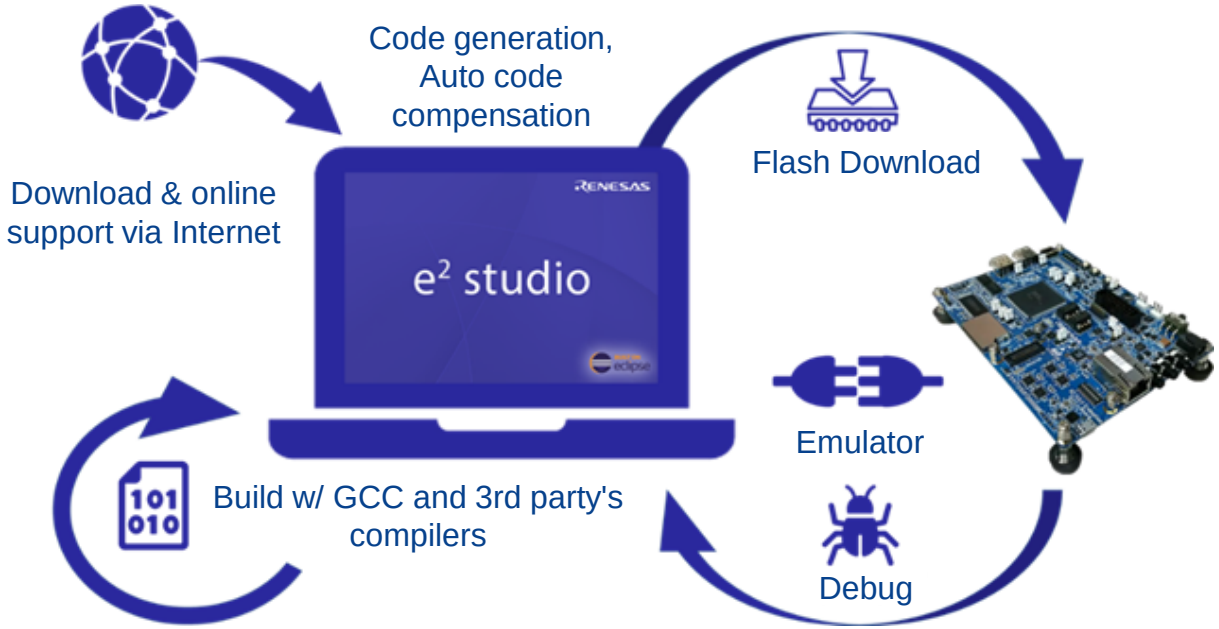
RTOS / FSP (Flexible Software Package)

IDE: e2studio

Based on the open source IDE "Eclipse" and the CDT (C/C++ Development Tooling) plug-in



All phases of Device selection, Pin configuration, Project generation, FSP drivers configuration, Build, debug and management are guided and supported by GUI

Integrate various functions for development

- ✓GNU compiler
- ✓Code editor
- ✓**Smart Configurator**
- ✓Smart manual

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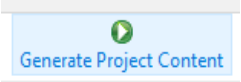
RTOS / FSP (Flexible Software Package)

SMART CONFIGURATOR

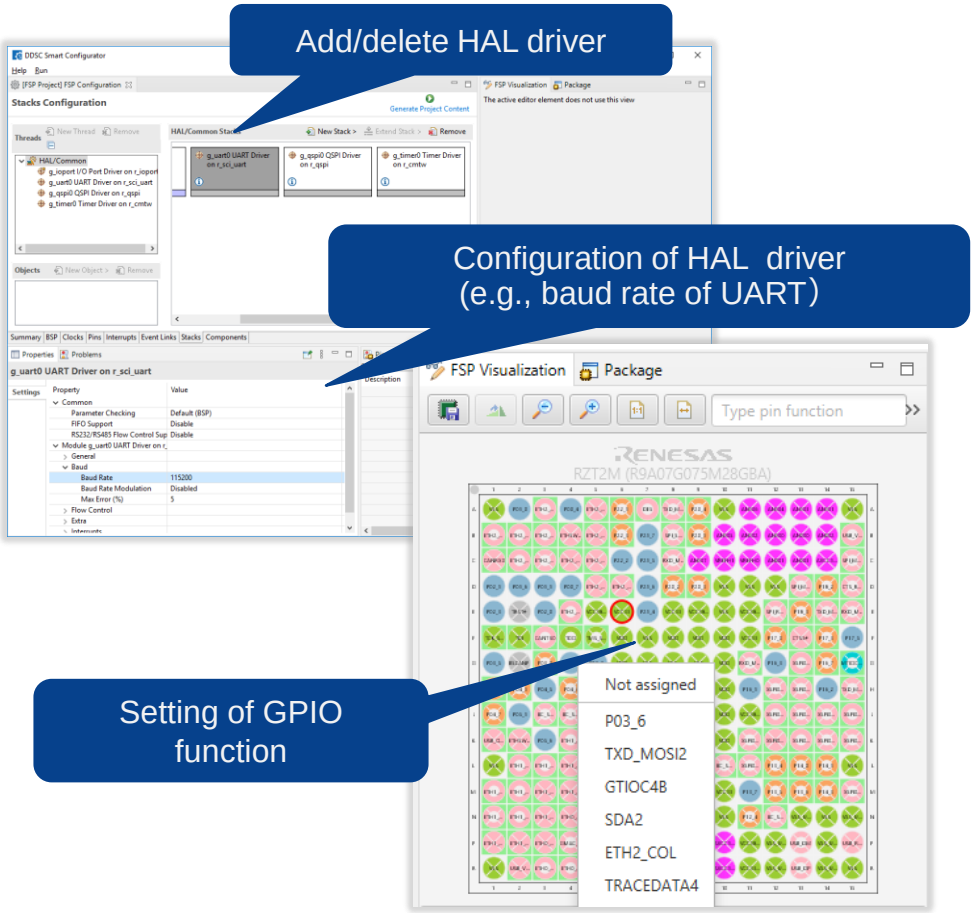
Various configurations are easily set by GUI

- ✓ Setting clock frequency
- ✓ Setting GPIO function
- ✓ Entry to interrupt handler
- ✓ Entry to event link
- ✓ Configuration of HAL driver
- ✓ Available tool(*) other than e2studio

After setting, code can be generated with one click and embedded in the user program



Configuration example



(*) for example, IAR Embedded Workbench for Arm

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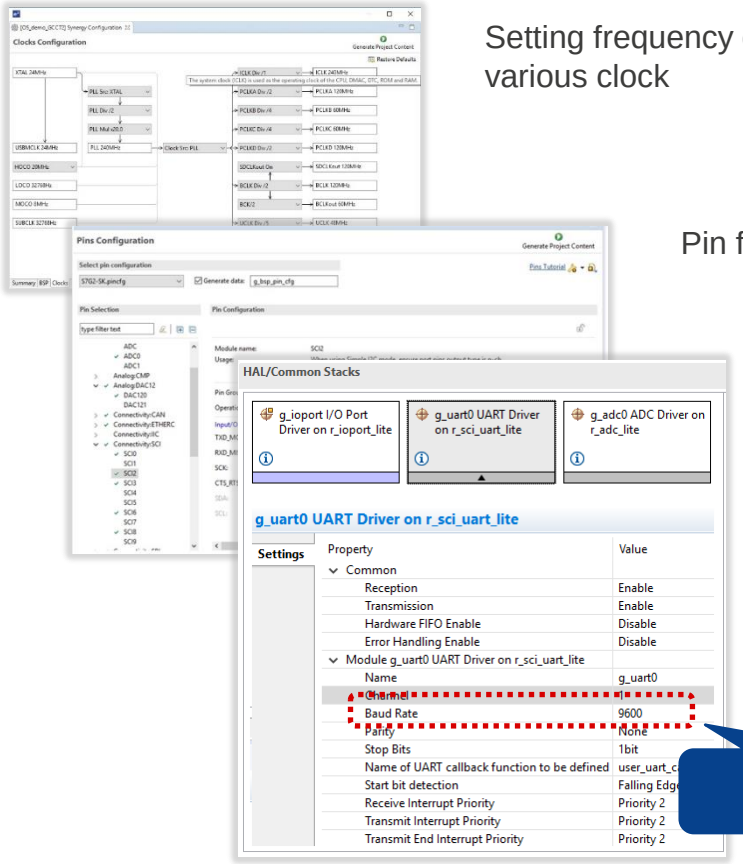
RTOS / FSP (Flexible Software Package)

SMART CONFIGURATOR

Code generation example



Setting frequency of various clock

Pin function setting

Configuration of HALdriver

Setting baud rate

Generate code

Example: hal_data.c

```
/* generated HAL source file - do not edit */
#include "hal_data.h"
#if (SCI_UART_CFG_RX_ENABLE)
#if (2) != BSP_IRQ_DISABLED
#if !defined(SSP_SUPPRESS_ISR_g_uart0) && !defined(SSP_SUPPRESS_ISR_SCI0)
SSP_VECTOR_DEFINE_CHAN(sci_uart_rxi_isr, SCI, RXI, 0);
#endif
#endif
#if (SCI_UART_ERROR_HANDLING_ENABLE)
#if (BSP_IRQ_DISABLED) != BSP_IRQ_DISABLED
#if !defined(SSP_SUPPRESS_ISR_g_uart0) && !defined(SSP_SUPPRESS_ISR_SCI0)
SSP_VECTOR_DEFINE_CHAN(sci_uart_eri_isr, SCI, ERI, 0);
#endif
#endif
#endif

sci_uart_instance_ctrl_t g_uart0_ctr

/** UART interface configuration */
const uart_cfg_t g_uart0_cfg =
{ .channel = 0, .baud_rate = 9600, .parity = UART_PARITY_OFF, .stop_bits
        user_uart_callback,
  .p_context = &g_uart0, .rx_edge_start = true, .rxi_ipl = (2), .txi_ipl
        (BSP_IRQ_DISABLED), };

/* Instance structure to use this module. */
const uart_instance_t g_uart0 =
{ .p_ctrl = &g_uart0_ctrl, .p_cfg = &g_uart0_cfg,
#if(BSP_CFG_INTERFACE_API_ENABLE)
  .p_api = &g_uart_on_sci_lite
#else
  .p_api = NULL
#endif
```

The set baud rate is reflected in the code

Generated code(include hal_data.c)are stored in following two folders

rzt_gen/

rzt_cfg/

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DEVELOPMENT ENVIRONMENT

RTOS / FSP

LINUX

EVALUATION BOARD

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LINUX SOLUTION OVERVIEW

CONTENTS OVERVIEW

- ✓ Linux Platform specialized for Industrial Automation
- ✓ Easy to start to develop industrial application on Linux Platform

RZ/T2N Linux Solution

Common Verified Linux PKG(VLP)

- GUI Framework (Qt, LVGL, etc.) (RZT2N not support)
- Linux standard libraries
- Linux kernel (CIP and CIP-RT)
- BSP (Device drivers)

+

Additional Packages

- Industrial Network (TSN, OPC UA, EtherCAT, etc.)
- Multi OS (OpenAMP, Cache partitioning)
- Security (Secure boot, ATF, OP-TEE, etc.)
- Software PLC (Codesys, OpenPLC, etc.)

[Note] RZ/T2N doesn't support GUI framework

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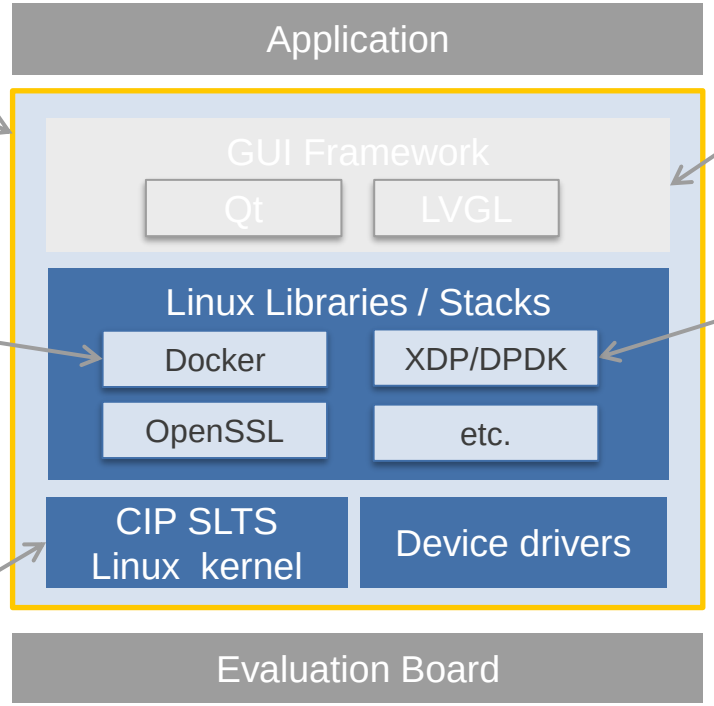
VERIFIED LINUX PACKAGE (VLP)

- ✓ Linux package verified by Renesas, including middleware
- ✓ Customer can start application development in stable environment

Verified Linux Package



Docker

・Easy to use container type virtualization

CIP SLTS Kernel

・ Civil Infrastructure Platform project
・10+ years super long term support
Reliability/Security/Real-time
(cf. LTS kernel: 2-6years support)

GUI Framework

(*RZ/T2N not support)

Data Plane Development Kit

- Accelerate network processing by not via kernel, but directly accessing network HW

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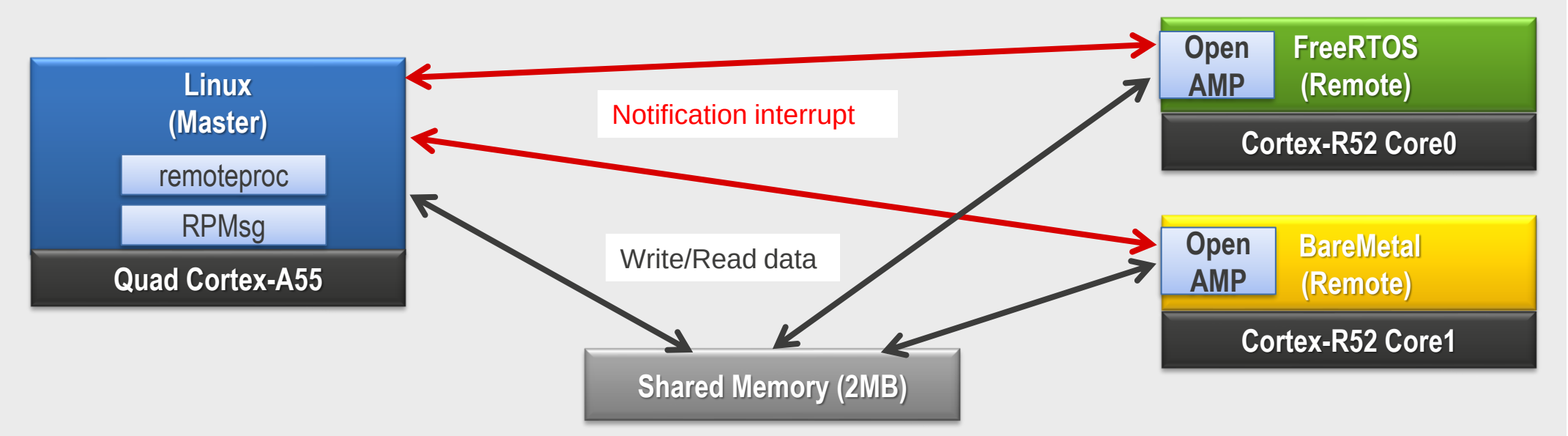
MULTI OS SOLUTION
CORE-TO-CORE COMMUNICATION

- ✓ Reference SW package for core-to-core communication for multi-OS co-operation
- ✓ Accelerates integration of Linux system and FreeRTOS/BareMetal systems

Multi OS package overview

- remoteproc and RPMsg are used in Linux, OpenAMP is used in FreeRTOS/BareMetal
- Linux and FreeRTOS/BareMetal communicate with notification interrupt and write/read data to/from shared memory



remoteproc (Remote Processor Framework): https://www.infradead.org/~mchehab/kernel_docs/staging/remoteproc.html
RPMsg (Remote Processor Messaging Framework): https://www.infradead.org/~mchehab/kernel_docs/staging/rpmsg.html

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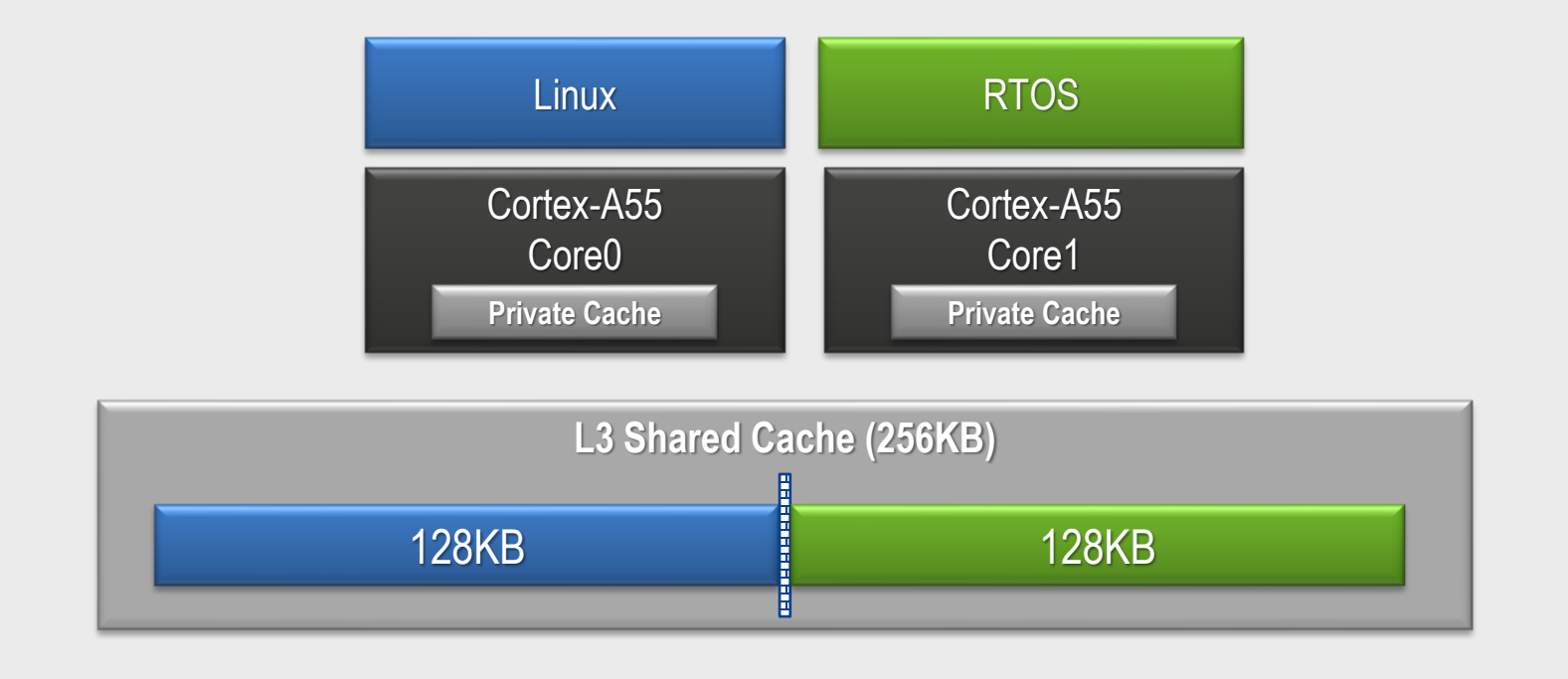
MULTI OS SOLUTION
SHARED CACHE PARTITIONING

- ✓ Provide not only Cortex-A55 – Cortex-R52 but also Cortex-A55 – Cortex-A55 sample
- ✓ Furthermore, provide shared cache partitioning sample program using Cortex-A55's DSU

For example, Controller application with strong real-time requirements can be decoupled from Linux by allocating Cortex-A55 resource as below



DSU: DynamIQ Shared Unit

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DEVELOPMENT ENVIRONMENT

RTOS / FSP

LINUX

EVALUATION BOARD

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EVALUATION BOARD FOR RZ/T2N
OVERVIEW

Features for various evaluation case

- ✓ For reference circuit, implement not only RZ/T2N 489-pin but also **power management IC DA9080** and GreenPAK
- ✓ Multi-axis motor evaluation by **Inverter board** option (max. 4-axis)
- ✓ On board multiplexer allows single board to support **5-port Gigabit Ethernet ports** or **4-axis motor control** or **external bus interface**
  External bus is for connecting external IC such as FPGA, ASIC, or memory
- ✓ Quick start debugging by **On-board emulator** (J-Link OB)
- ✓ Support **industrial voltage supply 24V** and generic USB-Type C PD supply 15V
- ✓ Various flash (OSPI flash, QSPI flash, eMMC, and NAND memory) and DRAM (LPDDR4, HyperRAM) for various application
- ✓ Firmware update is available by USB boot or SCI boot (*Need specific software)

No picture

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EVALUATION BOARD FOR RZ/T2N

SPECIFICATION

Item	Specification
CPU	RZ/T2N（R9A07G076M43GBA , 489-pin FBGA)
Memory	LPDDR4: 2GB HyperRAM: 8MB Octa Serial Flash: 64MB Quad Serial Flash: 16MB I2C EEPROM: 2KB eMMC: 32GB NAND: 4GB
Oscillator	For CPU: 25MHz For PCIe: 25MHz For USB serial IC: 12MHz
Power supply	15V/3A: USB PD Type-C / DC jack for AC adapter 24V/4A: 2-pole terminator
Switches	Push switch x4 Potentiometer (for ADC) x1
LED	Power 15V: yellow x1 Power 3.3V: green x1 User: green x6, yellow x1, red x2 Ethernet status (in RJ45): green x5, yellow x5 J-Link OB status: yellow x1

Item	Specification
Ethernet	RJ45 x 5 (through Gigabit PHY x5)
USB	Function: USB-MicroB Host: USB-TypeA On-the-Go: USB-MicroAB
USB-UART	USB-MicroB (through FT2232HQ)
SD card	microSD slot
CAN	3-pin 2.54 mm pitch (through CAN transceiver x1)
RS485	10-pin 2.54 mm pitch (through RS485 transceiver x1)
PCIe	1-lane x1 (Root Complex or Endpoint)
Debugger	J-Link OB: USB-MicroB MIPI-10 connector: 10-pin 1.27mm pitch MIPI-20 connector: 20-pin 1.27mm pitch
Pmod	Type2A/6A x1, Type3A x1
Pin header	External Bus: 40-pin 2.54mm pitch x2 Inverter Board connector: 50-pin 2.54mm pitch x4

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CONFIDENTIAL

REVISION HISTORY

Rev.	Date	Description
1.0	Sep 2024	Initial release

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